

Advanced (Carolina Reaper)

Q1. Simplify  $\sqrt{64}$

- (A) 8
- (B) 16
- (C) 32
- (D) 4
- (E) 2

- (A) 11
- (B) 12
- (C) 13
- (D) 14
- (E) 15

Q2. Simplify  $\sqrt[3]{27}$

- (A) 3
- (B) 9
- (C) 27
- (D) 1
- (E) 2

Q6. Simplify  $\sqrt[3]{125}$

- (A) 5
- (B) 10
- (C) 15
- (D) 20
- (E) 25

Q3. Convert to rational exponent form:  $\sqrt[4]{x}$

- (A)  $x^{1/2}$
- (B)  $x^{1/3}$
- (C)  $x^{1/4}$
- (D)  $x^{2/3}$
- (E)  $x^{3/4}$

Q7. A water tank can hold  $\sqrt[3]{1000}$  gallons of water. How many gallons can it hold?

- (A) 8
- (B) 10
- (C) 12
- (D) 9
- (E) 7

Q4. If  $a = \sqrt{b}$  and  $b = 16$ , what is  $a$ ?

- (A) 2
- (B) 4
- (C) 8
- (D) 16
- (E) 32

Q8. Tom is planning a road trip. The distance between two cities is  $\sqrt{256}$  miles. What is the distance?

- (A) 14
- (B) 16
- (C) 18
- (D) 20
- (E) 12

Q5. Simplify  $\sqrt{121}$

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Open Ended Questions (FRQ)

Q9. Simplify  $\sqrt[4]{x^2} \cdot \sqrt[3]{y^3}$  and convert to rational exponent form. Required work: 5 lines

Q10. A bakery sells  $\sqrt{100}$  cupcakes per day. If they make a profit of \$5 per cupcake, what is their total daily profit? Show all calculations. Required work: 5 lines

Chef's Tips

- Review all work carefully
- Use a calculator only when necessary
- Show all steps for free response questions